# Lecture 1

1/21/26
JRA

Welcome to Biomedical Imaging!

Today's lecture will present an overview of course mechanics & course's somewhat unconventional subject matter

## Course Mechanics - see Syllabus

- Subject matter
  - Microscopy
  - Medical Imaging
  - Image Processing & Analysis

- Office hours

- Grading

| | | |
|---|---|---|
| Homework | 30% | ok to work together |
| In-Class activities | 10% | |
| Midterm | 25% | - Microscopy |
| Final | 35% | Medical Imaging, Image Processing & Analysis |

## Subject Matter

Course is devoted to imaging

This is a hot topic driven by advances in computers & other technologies.

## (1) Light Microscopy

Light μscopy uses optical radiation to image specimens, living & dead

Electron μscopy uses high energy electrons to image specimens
- Higher resolution than light μscopy BUT
- Requires a vacuum ⇒ dead specimens only

Course covers whole spectrum of light μscopy techniques but does NOT cover electron μscopy

## (2) Medical Imaging

- Old days – did exploratory surgery
- Now – image

Imaging is a black box for most people

Knowing how image is formed is empowering, an avenue to great jobs, & helps to avoid mistakes. For example, in 3D microscopy, slice separation in the axial (z) direction should be ≤ ½ resolution to avoid sampling errors (like "aliasing") (e.g., 600 m resolution ⇒ sample at 30

## (3) Digital Image Processing & Analysis

"DIP" is used to clean up or rehabilitate images

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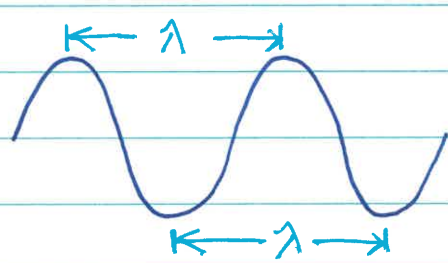
## Importance of Waves

Everything in class is based on interactions of waves w/ specimen or patient

Electromagnetic (EM) radiation is used w/ most techniques.

EM radiation is characterized by its wavelength, $\lambda$




$\lambda$ = peak-to-peak (or trough-to-trough) spacing

Differences in $\lambda$ give rise to markedly different properties

See initial PowerPoint slide

Decreasing Wavelength / Increasing Energy

Radio Waves - body mostly transparent to radio waves, so signal emerges to give information. Used in MRI.

Infrared (IR) - penetrate ~ 1 cm

Visible - penetrate a few mm
- Bad for imaging body
- Good for imaging cells

Ultraviolet (UV) - penetrates even less
- Kills cells
- Causes cataracts & skin cancer

X-Rays & Gamma Rays - penetrates body
- Some absorbed
- Some emerges to form image

- "ionizing radiation"

Ultrasound (US), in contrast, is a sound wave

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## Common Themes

There are several common themes that will run through the class

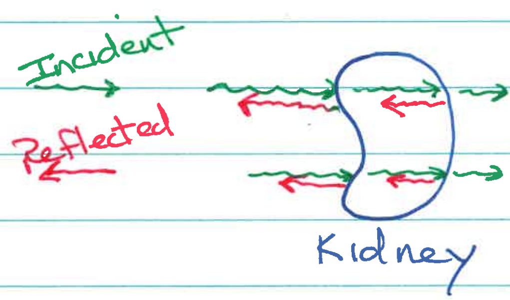
(1) Waves - waves interact w/ matter to form an image

(2) Basis (or mechanism) of image formation

We'll look at (1) how the particular wave (e.g., optical radiation, x-rays, ultrasound) interact w/ specimen/patient, & (2) how the interactions are used to create an image

Example - Ultrasound

Ultrasound uses reflected sound waves to form an image




- Send in incident sound waves
- Detect reflected sound waves (i.e., detect "echoes")

- Echo Time - monitor echo time & use to determine object depth
  - Surface structure - time for echo to return is short
  - Deep structures - time for echo to return is long

- Echo Strength – monitor echo strength to determine object brightness

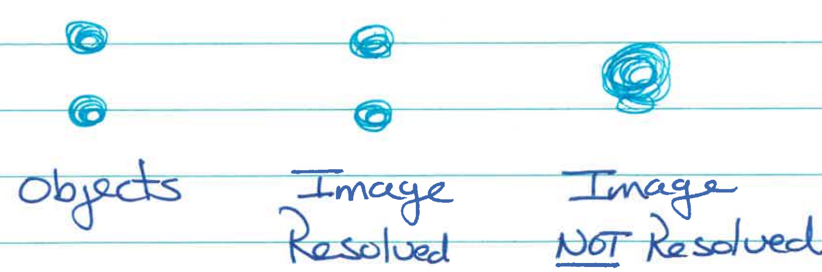
  - Strong echo – bright
  - Weak echo – dim

## (3) Image Quality

Look at several measures to charactize image quality, especially resolution & contrast

### (3a) Spatial Resolution

Spatial resolution refers to the ability to distinguish objects as distinct, versus having them overlap in image & look like one object




How close can objects get before they cannot be distinguished?
How close & still create distinct images?

- Standard light μscopy

Resolution limit is ~ 200 nm (0.2 μm)

Severe limitation – objects closer than this cannot be distinguished

Cells are ~ 10 μm in size, so resolution is ~ 2% of cell size

In the late 1990s, biophysicists developed superresolution light μscopy
Resolution ≈ 20 nm

- Medical imaging

Resolution is a much bigger number than w/ μscopy, but the body is much larger than a cell, so relative resolution's more comparable

Resolution limit is ~ 0.1 mm to ~ 1 cm
↑ x-ray ↑ nuclear imaging

These techniques all have strengths & weaknesses

## (3b) Contrast

Contrast provides a way to distinguish signal variations at different points in an image

- Grayscale variations – differences in white, gray, black
- Color variations – differences in color (& intensity)

Otherwise, w/o contrast, can't see anything

See PowerPoint slide w/ nerve cell images generated w/ different transmitted light microscopy techniques (all grayscale)

- Brightfield – cells almost invisible
- Darkfield – cells bright on dark background
- Phase contrast – cells dark on bright background

Fluorescence microscopy uses color for contrast

MRI is a powerful technique not because of its resolution, which is standard, but because of its exquisite soft tissue contrast

## (3c) Temporal Resolution

How quickly can you collect images?

- Ultrasound – excellent (real-time) temporal resolution (~30 frames/sec) (can see beating heart)
- MRI – poor temporal resolution (very slow)

## (3d) Noise

Noise is random signal superimposed on the signal of interest

Nuclear images (generated by collecting radioactivity) are very prone to being noisy

## (3e) Structural vs Functional

Does technique yield structural information or functional information?

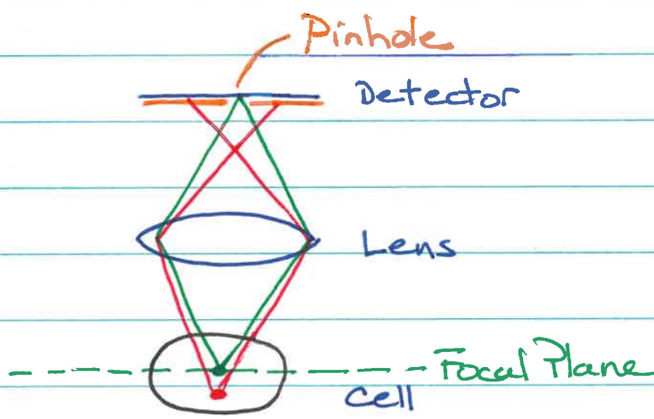
- Structural – is rib broken?
- Functional – is thyroid hypo or hyper active?

## Role of Physics

First-year physics allows an understanding of a lot

<u>Example 1</u> - how does confocal microscopy deblur images?

Confocal μscopy was the brainchild of Marvin Minsky (of artificial intelligence fame) in the 1950s




Green light, from focal plane, in focus at detector & passes thru pinhole

Red light, from below focal plane, in focus before detector & blocked because it misses pinhole

This is why standard fluorescence images are blurry: they are formed from fluorescence originating above & below the focal plane, as well as in the focal plane, contaminating the image

Confocal μscopy

- In-focus light makes it thru pinhole & is captured by detector
- Out-of-focus light is mostly blocked at pinhole (it's "rejected") & so is <u>NOT</u> captured by detector

<u>Example 2</u> - how are ultrasound images formed?

Looked at ultrasound simulation showing that imaging requires reflection <u>AND</u> transmission at interfaces (the latter to allow imaging of deeper structures)